CYBER SECURITY

CAPTURE THE FLAG (CTF) MACHINE REPORT

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SRN:R25DE075

1. Title Page

Title: Development of a Capture The Flag (CTF) Machine on Linux

2. Objective

The objective of this project is to design and deploy a Capture The Flag (CTF) environment on Kali Linux that simulates real-world penetration testing scenarios. The system includes web enumeration, steganography, credential discovery, SSH access, and privilege separation leading to flag retrieval.

3. Scope of the Project

This project covers Linux user management, web server configuration using Apache2, hidden information disclosure techniques, steganography using steghide, and secure shell access via SSH.

4. Tools and Technologies Used

- Kali Linux Rolling Edition
- Apache2 Web Server
- OpenSSH Server
- Steghide Tool
- Linux user management utilities
- Nano text editor

5. System Architecture

The system is structured into multiple layers: Web Layer (Apache), File System Layer (hidden files and stego), and System Layer (Linux users and SSH access).

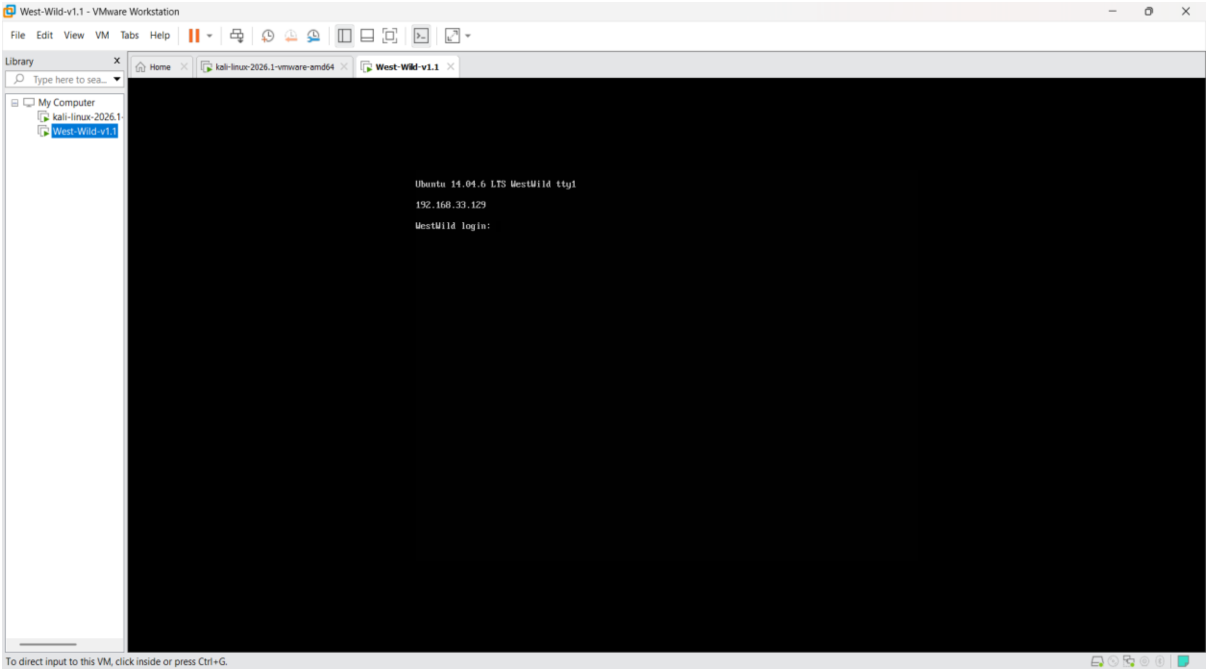
6. Network Configuration

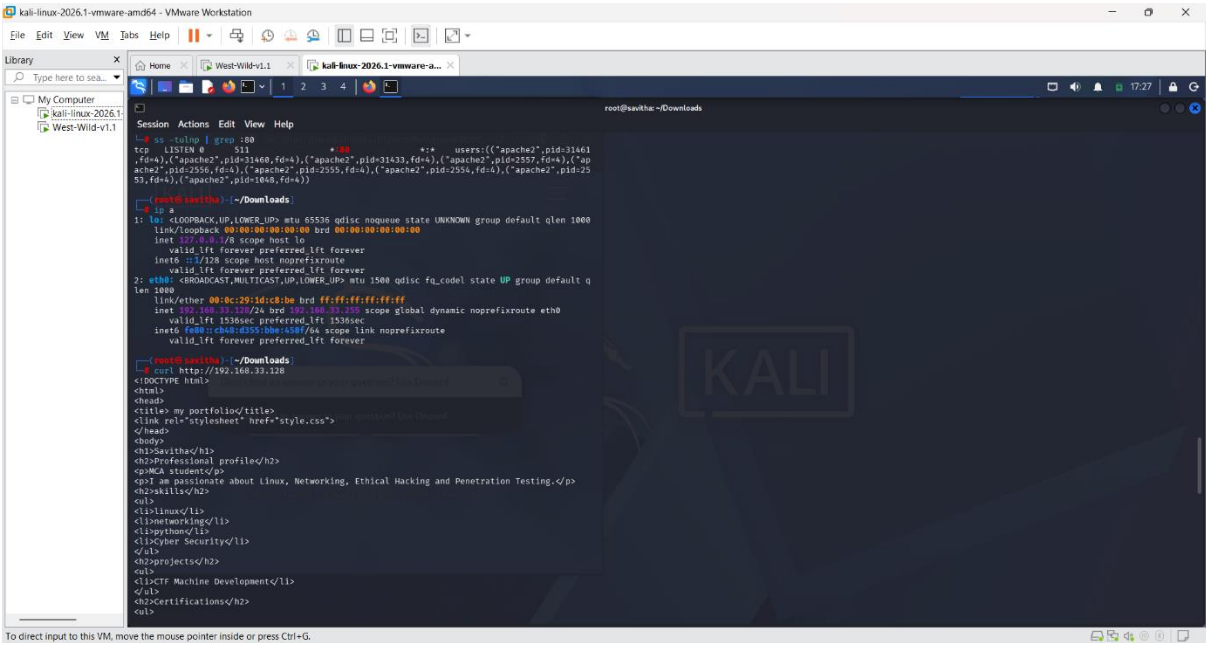
IP Address: 192.168.33.128

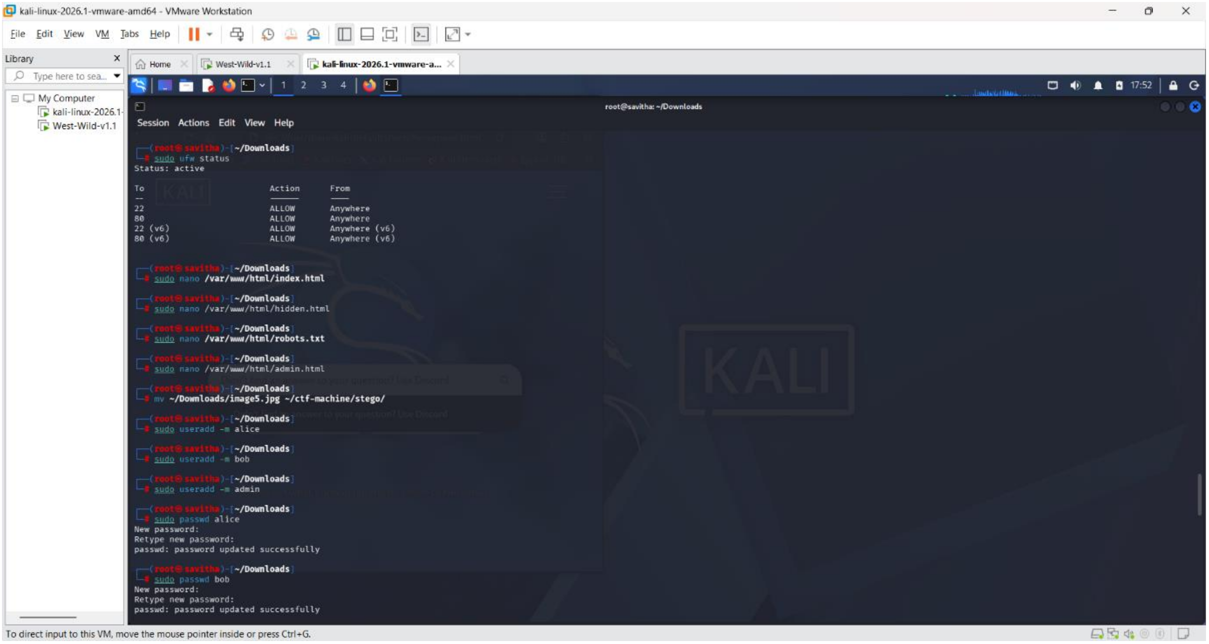
Subnet: 255.255.255.0

Ports Open: 22 (SSH), 80 (HTTP)

Network Mode: NAT/Bridged (VM dependent)



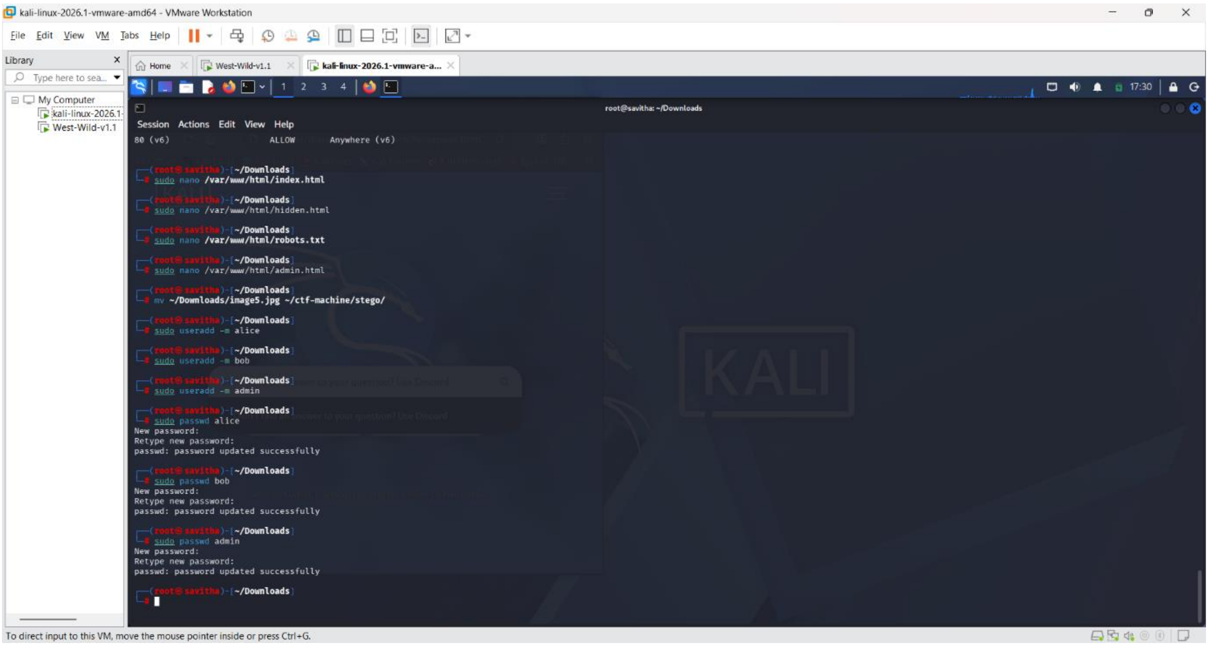




7. User Accounts Created

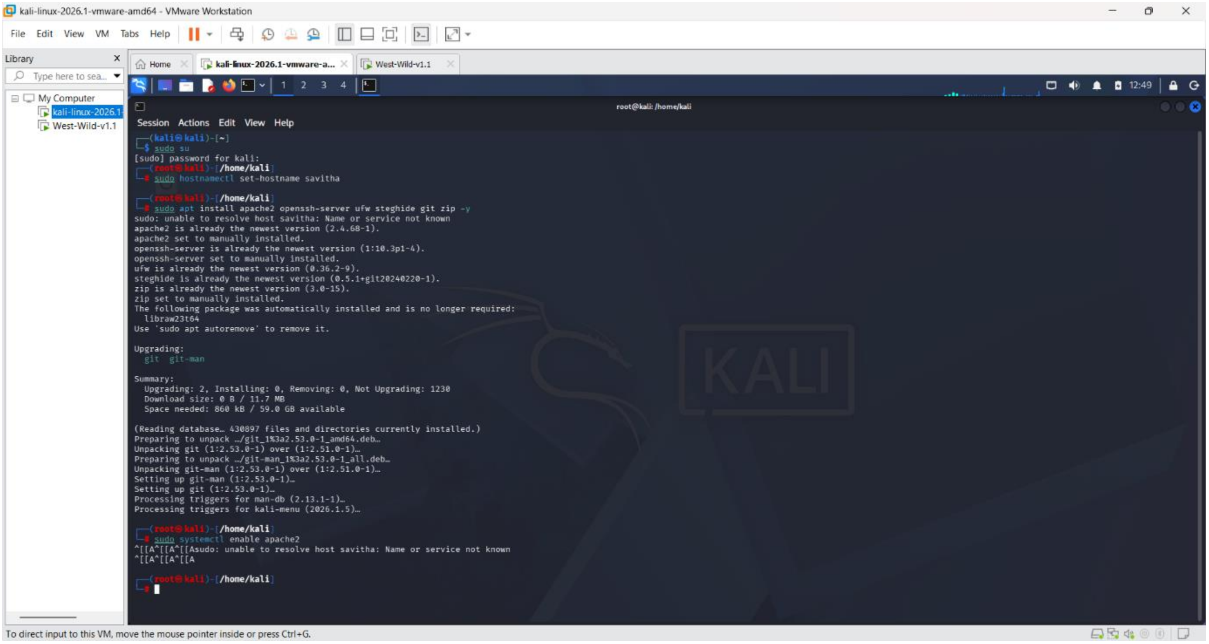
Three system users were created for the challenge:

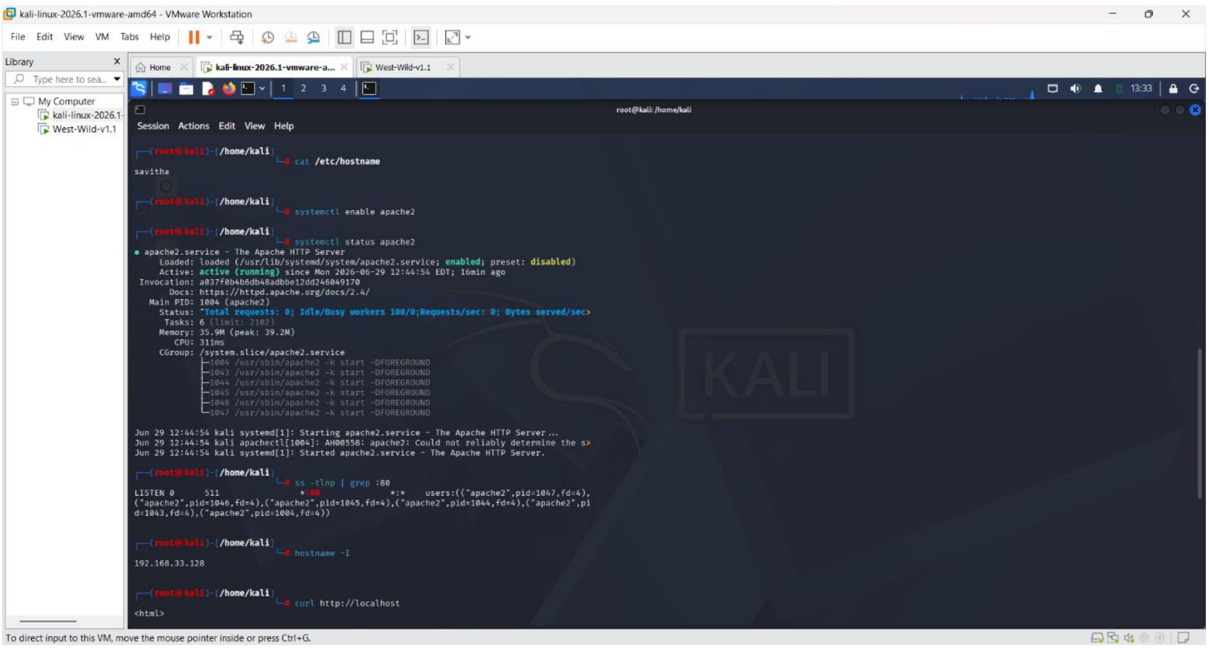
- alice (primary target user)
- bob (decoy user)
- admin (privileged user)

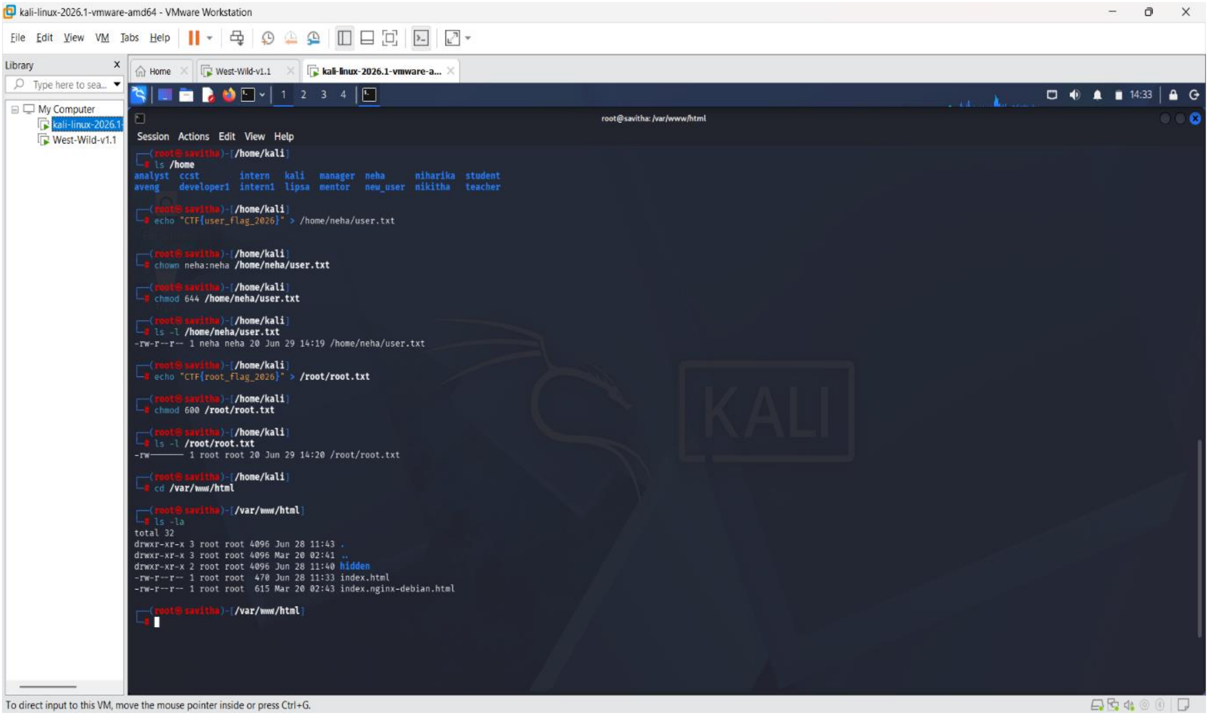


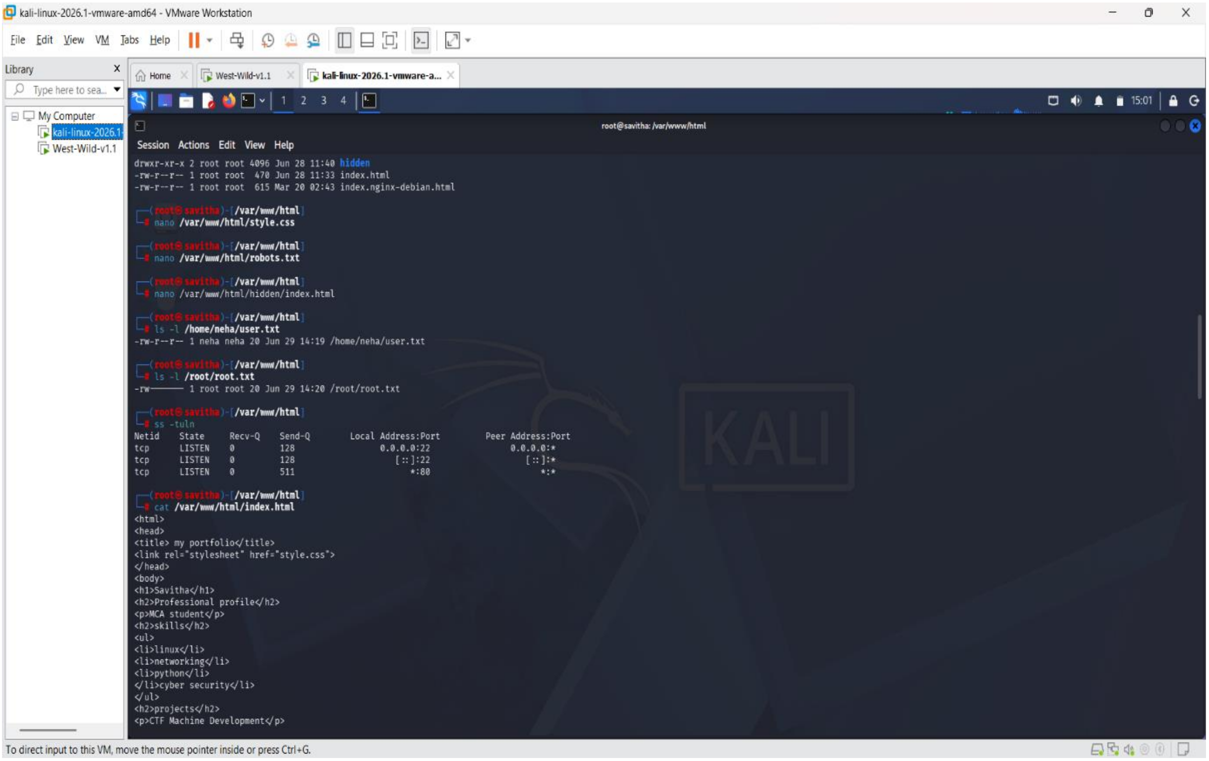
8. Web Application Setup

A custom Apache2 web page was developed containing hidden clues in HTML comments, hidden pages, and enumeration hints such as robots.txt and fake admin panels.



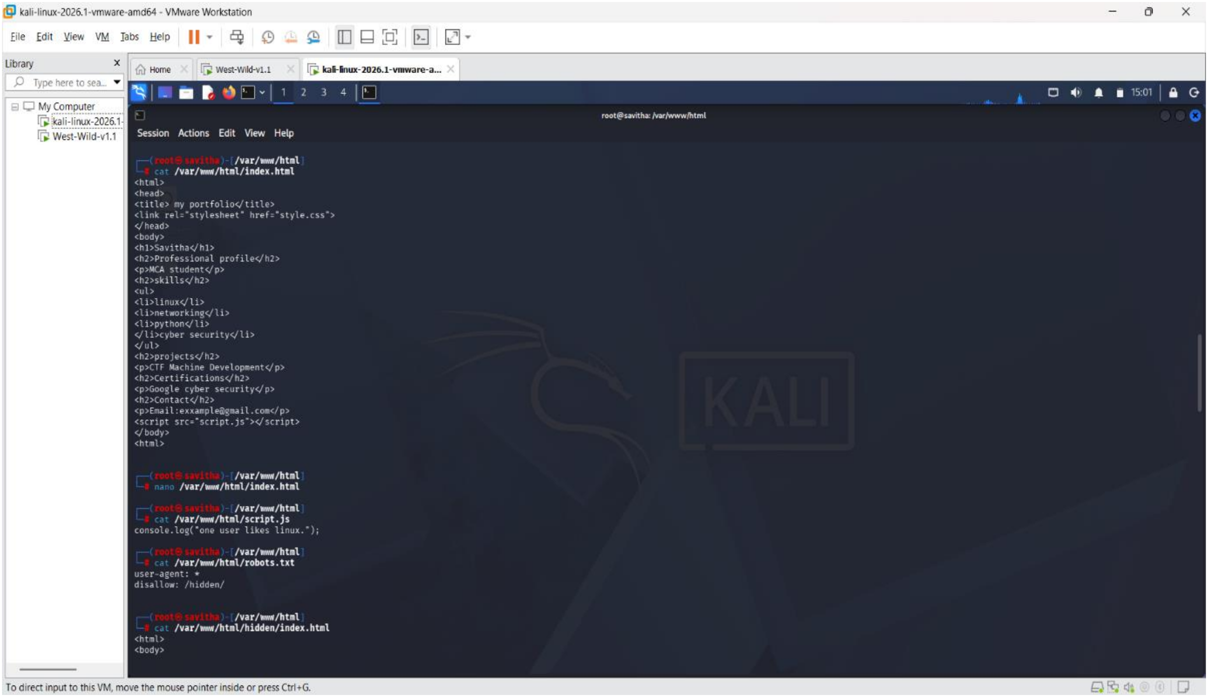



kali-linux-2026.1-vmware-amd64 - VMware Workstation
root@savitha: /var/www/html
Session Actions Edit View Help
┌──(root㉿savitha)-[/home/kali]
└─# ls /home
analyst ccst intern kali manager neha niharika student
aveng developer1 intern1 lipsa mentor new_user nikitha teacher
┌──(root㉿savitha)-[/home/kali]
└─# echo "CTF{user_flag_2026}" > /home/neha/user.txt
┌──(root㉿savitha)-[/home/kali]
└─# chown neha:neha /home/neha/user.txt
┌──(root㉿savitha)-[/home/kali]
└─# chmod 644 /home/neha/user.txt
┌──(root㉿savitha)-[/home/kali]
└─# ls -l /home/neha/user.txt
-rw-r--r-- 1 neha neha 20 Jun 29 14:19 /home/neha/user.txt
┌──(root㉿savitha)-[/home/kali]
└─# echo "CTF{root_flag_2026}" > /root/root.txt
┌──(root㉿savitha)-[/home/kali]
└─# chmod 600 /root/root.txt
┌──(root㉿savitha)-[/home/kali]
└─# ls -l /root/root.txt
-rw------- 1 root root 20 Jun 29 14:20 /root/root.txt
┌──(root㉿savitha)-[/home/kali]
└─# cd /var/www/html
┌──(root㉿savitha)-[/var/www/html]
└─# ls -la
total 32
drwxr-xr-x 3 root root 4096 Jun 28 11:43 .
drwxr-xr-x 3 root root 4096 Mar 20 02:41 ..
drwxr-xr-x 2 root root 4096 Jun 28 11:40 hidden
-rw-r--r-- 1 root root 470 Jun 28 11:33 index.html
-rw-r--r-- 1 root root 615 Mar 20 02:43 index.nginx-debian.html
┌──(root㉿savitha)-[/var/www/html]
└─#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2026.1-vmware-amd64 - VMware Workstation
root@savitha: /var/www/html
Session Actions Edit View Help
drwxr-xr-x 2 root root 4096 Jun 28 11:40 hidden
-rw-r--r-- 1 root root 470 Jun 28 11:33 index.html
-rw-r--r-- 1 root root 615 Mar 20 02:43 index.nginx-debian.html
┌──(root㉿savitha)-[/var/www/html]
└─# nano /var/www/html/style.css
┌──(root㉿savitha)-[/var/www/html]
└─# nano /var/www/html/robots.txt
┌──(root㉿savitha)-[/var/www/html]
└─# nano /var/www/html/hidden/index.html
┌──(root㉿savitha)-[/var/www/html]
└─# ls -l /home/neha/user.txt
-rw-r--r-- 1 neha neha 20 Jun 29 14:19 /home/neha/user.txt
┌──(root㉿savitha)-[/var/www/html]
└─# ls -l /root/root.txt
-rw------- 1 root root 20 Jun 29 14:20 /root/root.txt
┌──(root㉿savitha)-[/var/www/html]
└─# ss -tuln
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:*
tcp LISTEN 0 128 [::]:22 [::]:*
tcp LISTEN 0 511 *:80 *:*
┌──(root㉿savitha)-[/var/www/html]
└─# cat /var/www/html/index.html
<html>
<head>
<title> my portfolio</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Savitha</h1>
<h2>Professional profile</h2>
<p>MCA student</p>
<h2>skills</h2>

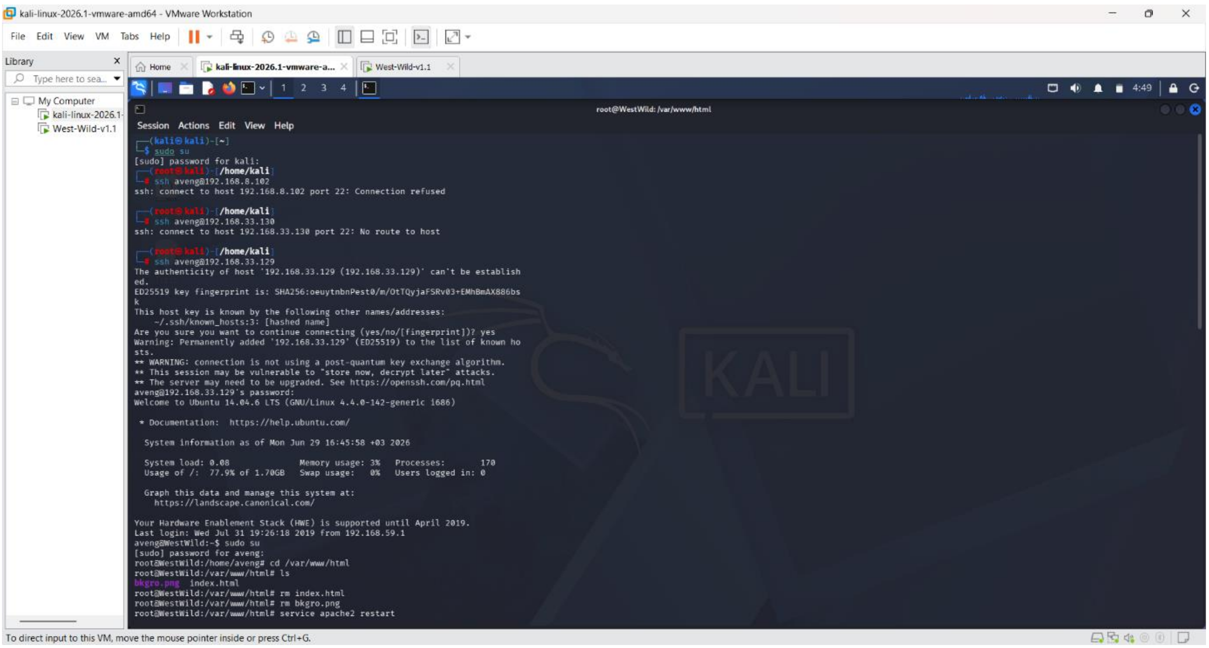
linux
networking
python
cyber security

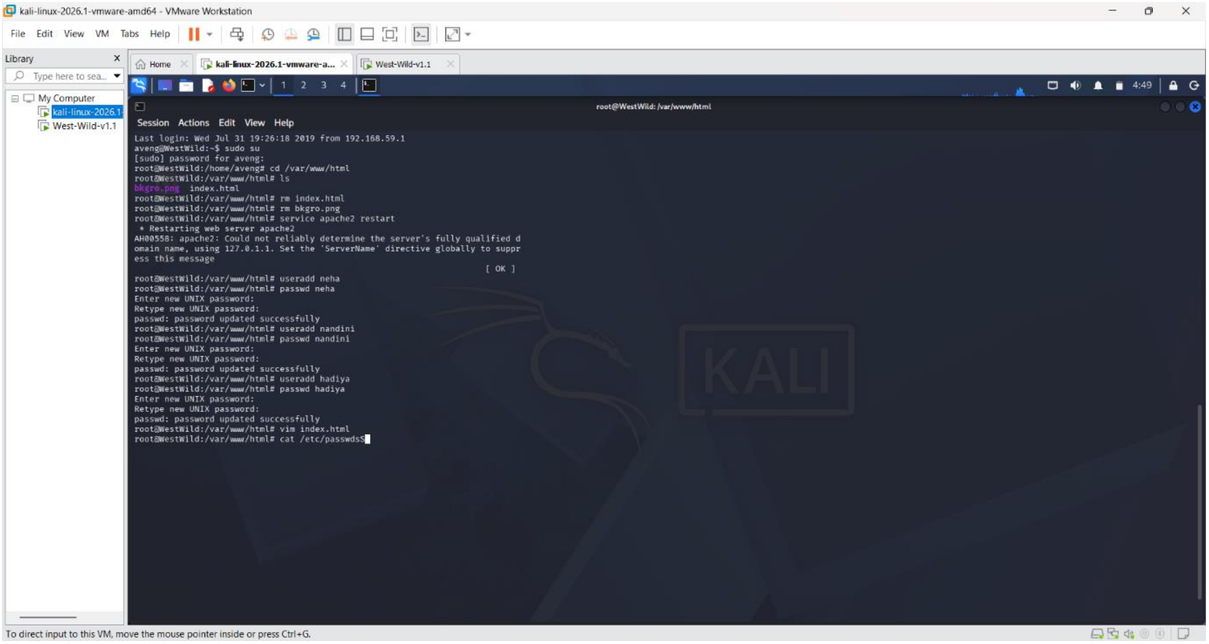
<h2>projects</h2>
<p>CTF Machine Development</p>
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



9. Hidden Information Disclosure

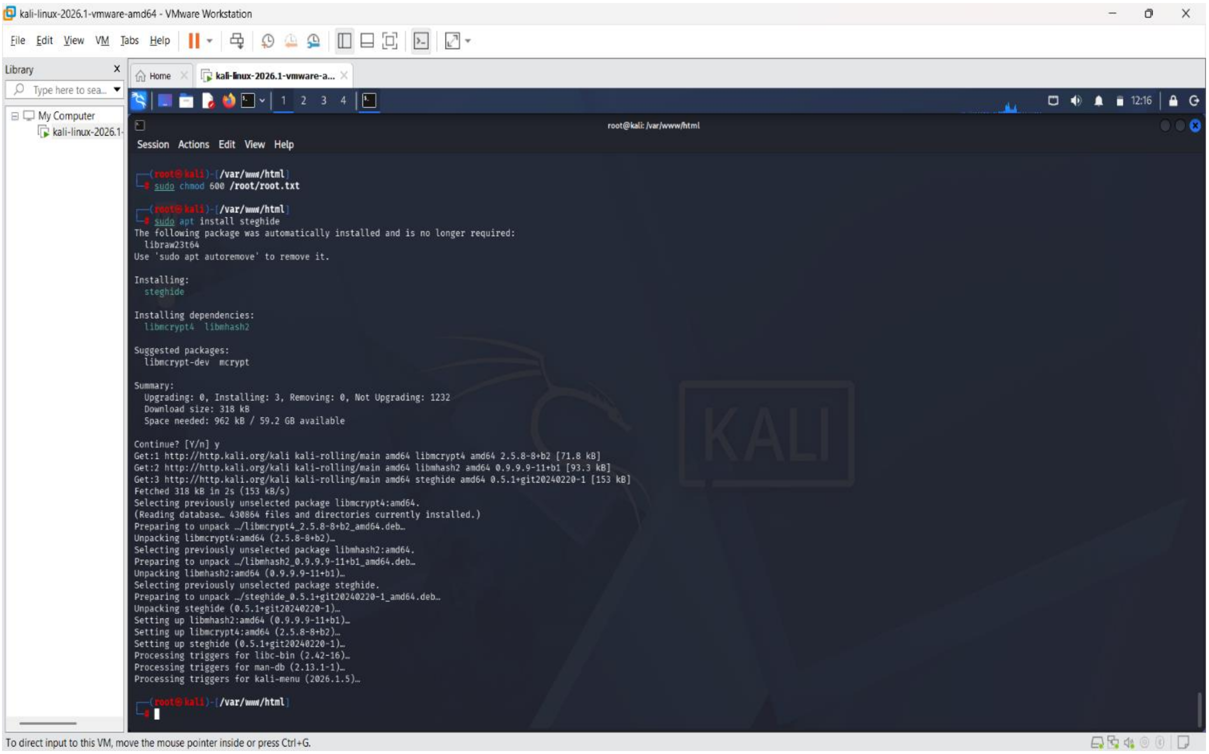
Sensitive hints were embedded using HTML comments, hidden directories, and intentionally exposed files to simulate misconfiguration vulnerabilities.

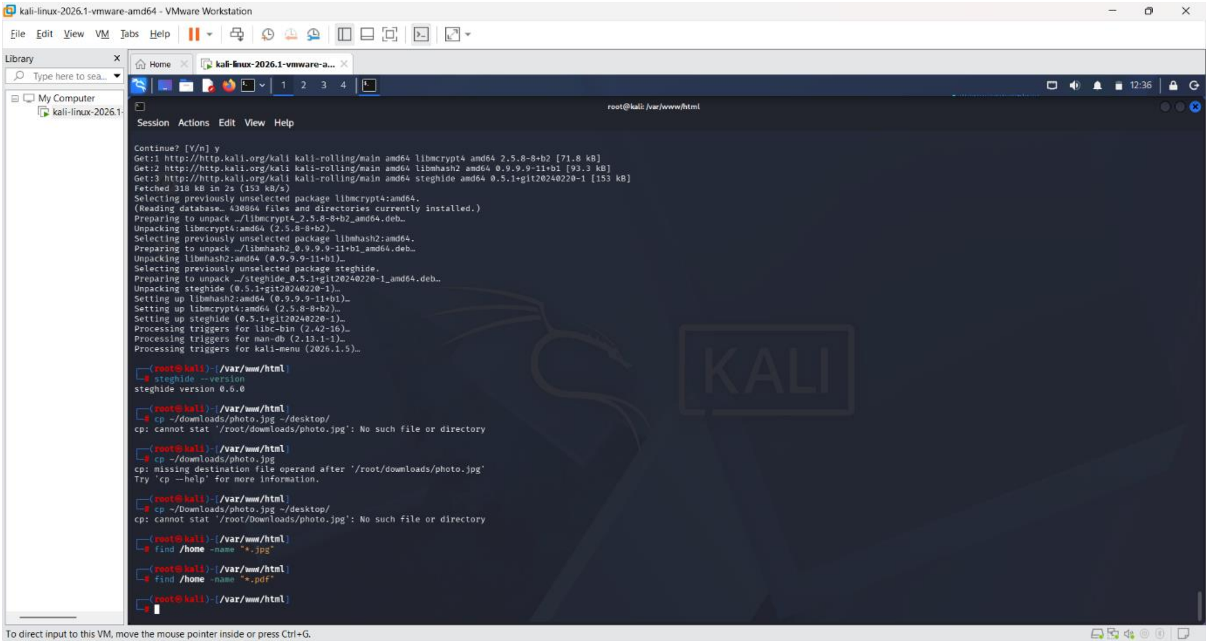


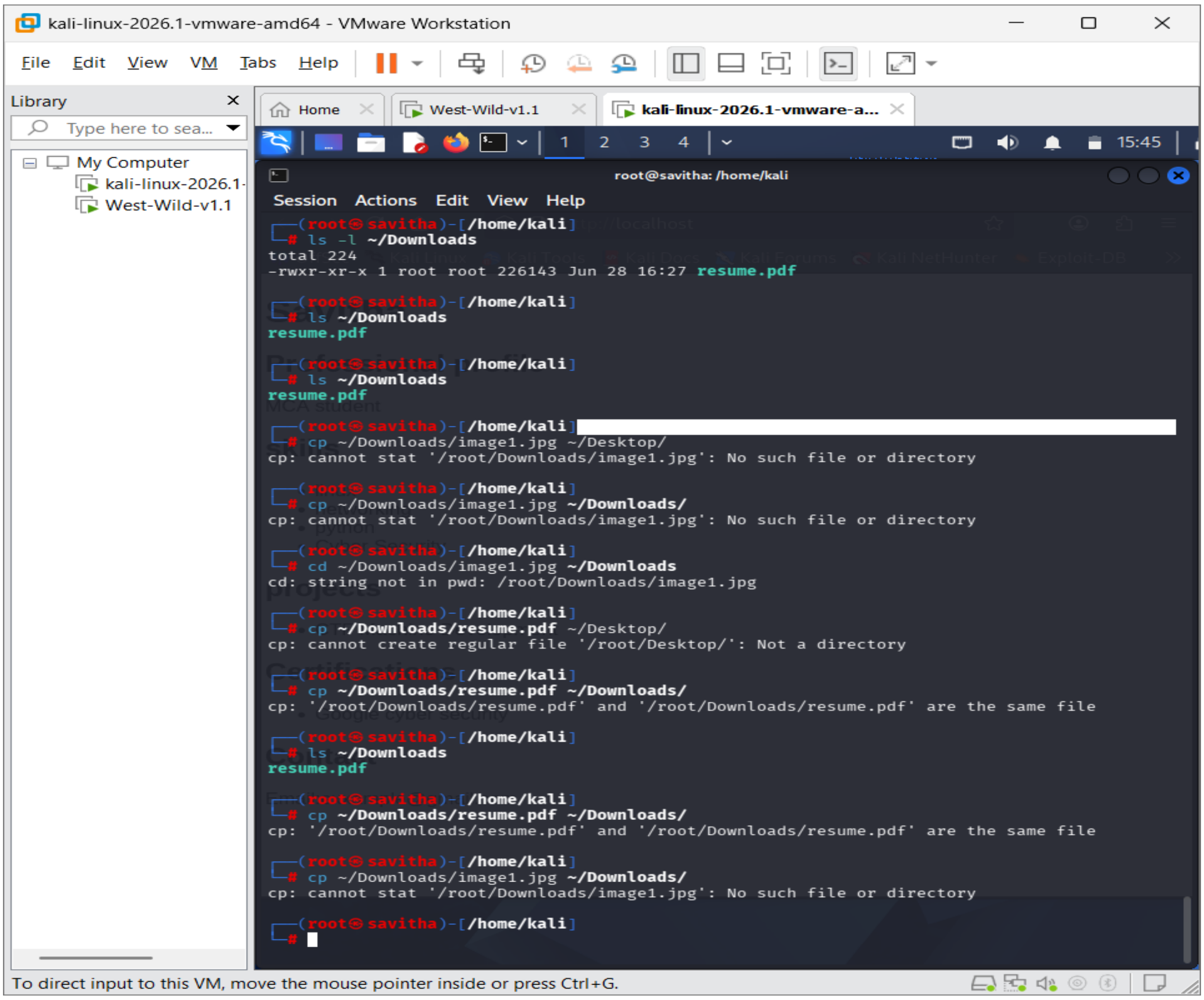


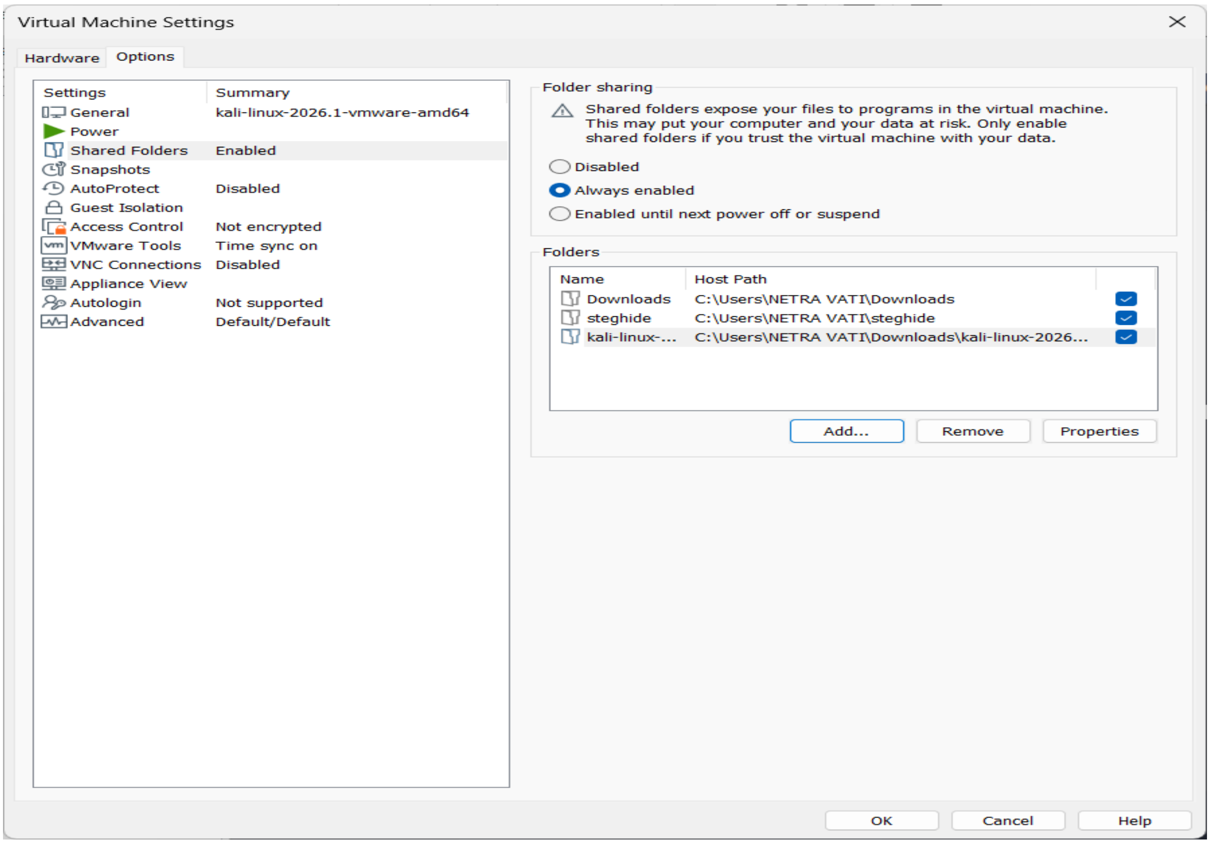
10. Steganography Implementation

An image file (image5.jpg) was used to hide sensitive data using steghide. The extracted file secret.txt contained login credentials required for SSH access.

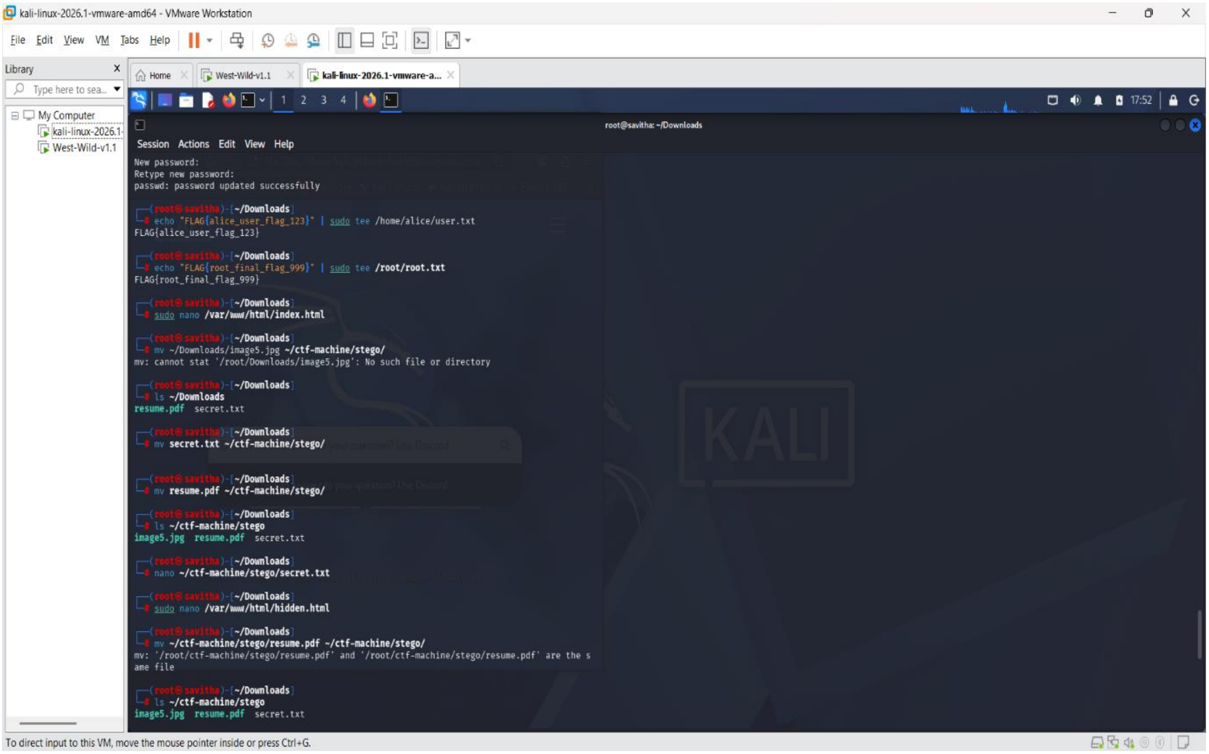



kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Home
kali-linux-2026.1-vmware-a...
My Computer
kali-linux-2026.1-
12:36
root@kali: /var/www/html
Session Actions Edit View Help
Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libmcrypt4 amd64 2.5.8-8+b2 [71.8 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libmhash2 amd64 0.9.9.9-11+b1 [93.3 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 steghide amd64 0.5.1+git20240220-1 [153 kB]
Fetched 318 kB in 2s (153 kB/s)
Selecting previously unselected package libmcrypt4:amd64.
(Reading database… 430864 files and directories currently installed.)
Preparing to unpack …/libmcrypt4_2.5.8-8+b2_amd64.deb…
Unpacking libmcrypt4:amd64 (2.5.8-8+b2)…
Selecting previously unselected package libmhash2:amd64.
Preparing to unpack …/libmhash2_0.9.9.9-11+b1_amd64.deb…
Unpacking libmhash2:amd64 (0.9.9.9-11+b1)…
Selecting previously unselected package steghide.
Preparing to unpack …/steghide_0.5.1+git20240220-1_amd64.deb…
Unpacking steghide (0.5.1+git20240220-1)…
Setting up libmhash2:amd64 (0.9.9.9-11+b1)…
Setting up libmcrypt4:amd64 (2.5.8-8+b2)…
Setting up steghide (0.5.1+git20240220-1)…
Processing triggers for libc-bin (2.42-16)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.1.5)…
(root㉿kali)-[/var/www/html]
steghide --version
steghide version 0.6.0
(root㉿kali)-[/var/www/html]
cp ~/downloads/photo.jpg ~/desktop/
cp: cannot stat '/root/downloads/photo.jpg': No such file or directory
(root㉿kali)-[/var/www/html]
cp ~/downloads/photo.jpg
cp: missing destination file operand after '/root/downloads/photo.jpg'
Try 'cp --help' for more information.
(root㉿kali)-[/var/www/html]
cp ~/Downloads/photo.jpg ~/desktop/
cp: cannot stat '/root/Downloads/photo.jpg': No such file or directory
(root㉿kali)-[/var/www/html]
find /home -name "*.jpg"
(root㉿kali)-[/var/www/html]
find /home -name "*.pdf"
(root㉿kali)-[/var/www/html]
#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

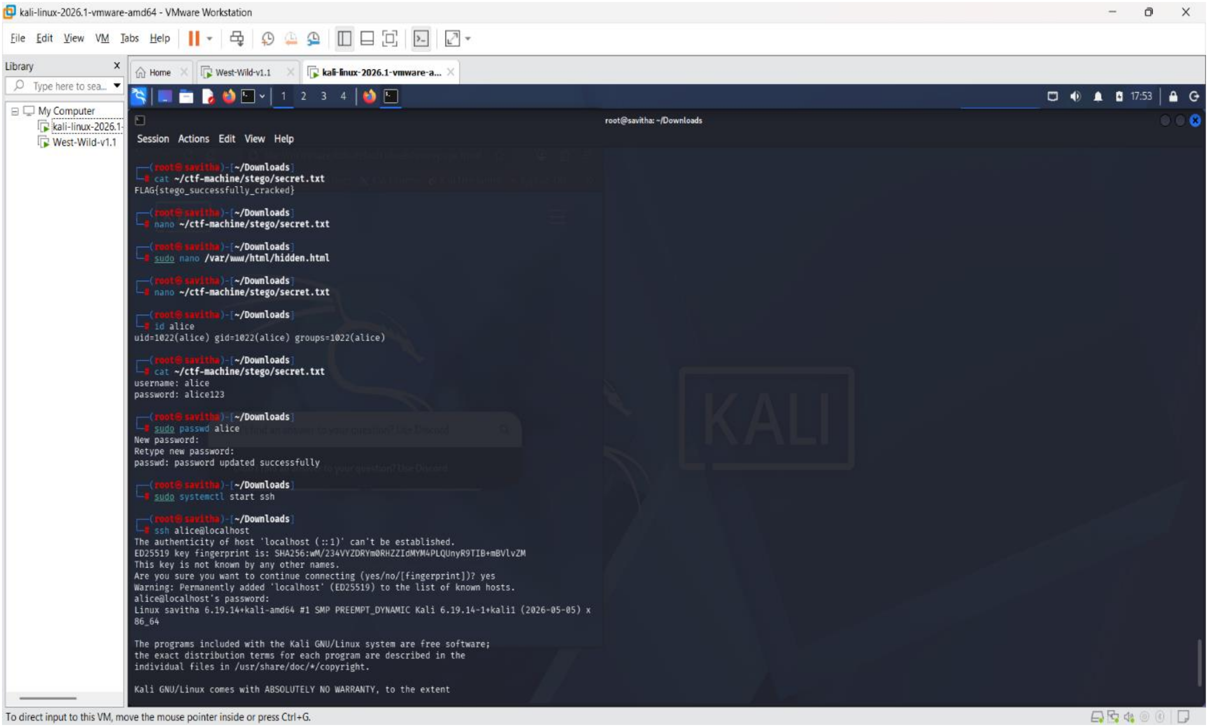

kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
kali-linux-2026.1-
West-Wild-v1.1
Home
West-Wild-v1.1
kali-linux-2026.1-vmware-a...
1 2 3 4
15:45
root@savitha: /home/kali
Session Actions Edit View Help
(root㉿savitha)-[/home/kali]
ls -l ~/Downloads
total 224
-rwxr-xr-x 1 root root 226143 Jun 28 16:27 resume.pdf
(root㉿savitha)-[/home/kali]
ls ~/Downloads
resume.pdf
(root㉿savitha)-[/home/kali]
ls ~/Downloads
resume.pdf
(root㉿savitha)-[/home/kali]
cp ~/Downloads/image1.jpg ~/Desktop/
cp: cannot stat '/root/Downloads/image1.jpg': No such file or directory
(root㉿savitha)-[/home/kali]
cp ~/Downloads/image1.jpg ~/Downloads/
cp: cannot stat '/root/Downloads/image1.jpg': No such file or directory
(root㉿savitha)-[/home/kali]
cd ~/Downloads/image1.jpg ~/Downloads
cd: string not in pwd: /root/Downloads/image1.jpg
(root㉿savitha)-[/home/kali]
cp ~/Downloads/resume.pdf ~/Desktop/
cp: cannot create regular file '/root/Desktop/': Not a directory
(root㉿savitha)-[/home/kali]
cp ~/Downloads/resume.pdf ~/Downloads/
cp: '/root/Downloads/resume.pdf' and '/root/Downloads/resume.pdf' are the same file
(root㉿savitha)-[/home/kali]
ls ~/Downloads
resume.pdf
(root㉿savitha)-[/home/kali]
cp ~/Downloads/resume.pdf ~/Downloads/
cp: '/root/Downloads/resume.pdf' and '/root/Downloads/resume.pdf' are the same file
(root㉿savitha)-[/home/kali]
cp ~/Downloads/image1.jpg ~/Downloads/
cp: cannot stat '/root/Downloads/image1.jpg': No such file or directory
(root㉿savitha)-[/home/kali]
#
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



Virtual Machine Settings
Hardware
Options
Settings
Summary
General
kali-linux-2026.1-vmware-amd64
Power
Shared Folders
Enabled
Snapshots
AutoProtect
Disabled
Guest Isolation
Access Control
Not encrypted
VMware Tools
Time sync on
VNC Connections
Disabled
Appliance View
Autologin
Not supported
Advanced
Default/Default
Folder sharing
Shared folders expose your files to programs in the virtual machine. This may put your computer and your data at risk. Only enable shared folders if you trust the virtual machine with your data.
Disabled
Always enabled
Enabled until next power off or suspend
Folders
Name
Host Path
Downloads
C:\Users\NETRA VATI\Downloads
steghide
C:\Users\NETRA VATI\steghide
kali-linux-...
C:\Users\NETRA VATI\Downloads\kali-linux-2026...
Add...
Remove
Properties
OK
Cancel
Help

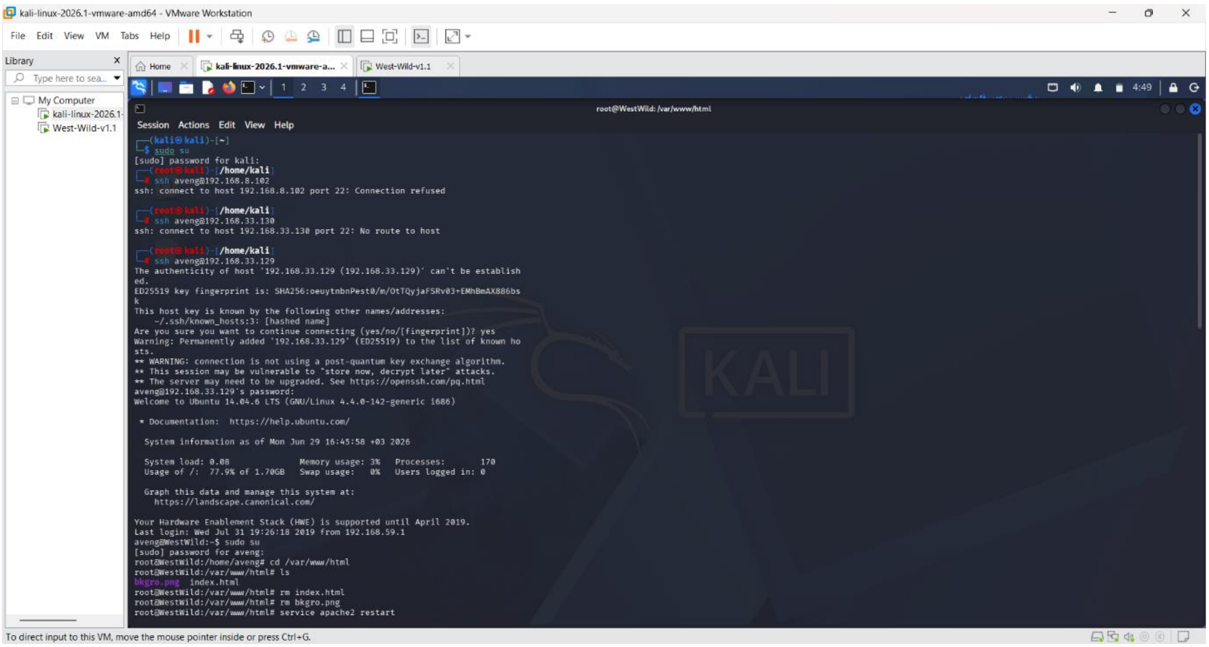


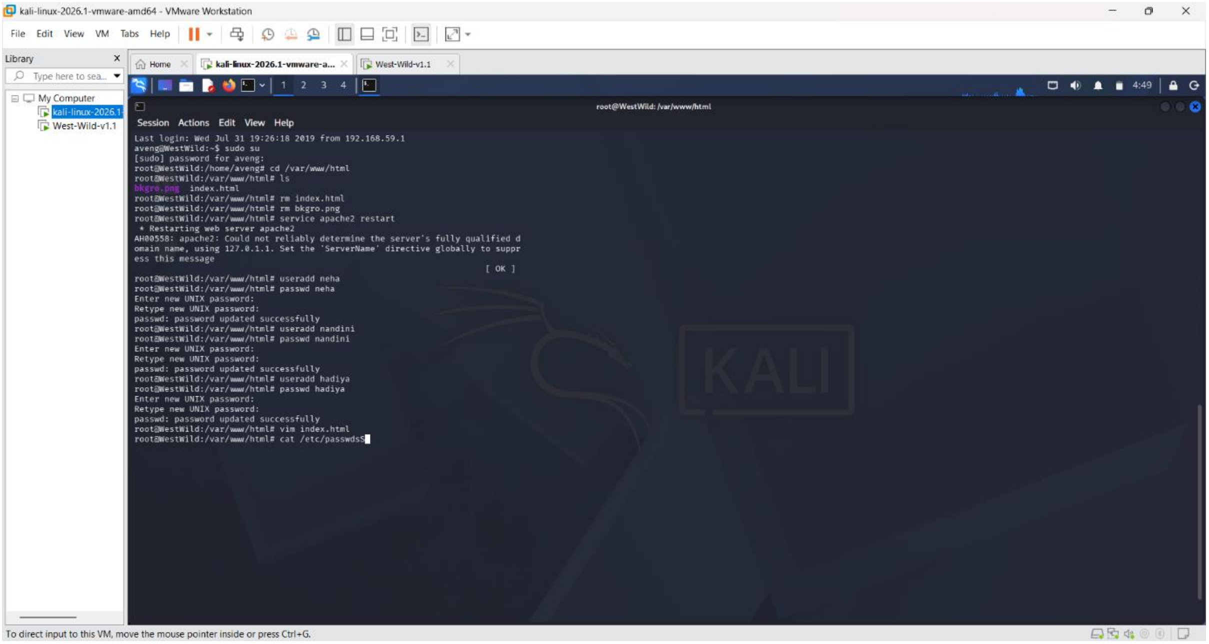
kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Home
West-Wild-v1.1
kali-linux-2026.1-vmware-a...
My Computer
kali-linux-2026.1
West-Wild-v1.1
root@savitha: ~/Downloads
Session Actions Edit View Help
New password:
Retype new password:
passwd: password updated successfully
echo "FLAG{alice_user_flag_123}" | sudo tee /home/alice/user.txt
FLAG{alice_user_flag_123}
echo "FLAG{root_final_flag_999}" | sudo tee /root/root.txt
FLAG{root_final_flag_999}
sudo nano /var/www/html/index.html
mv ~/Downloads/image5.jpg ~/ctf-machine/stego/
mv: cannot stat '/root/Downloads/image5.jpg': No such file or directory
ls ~/Downloads
resume.pdf secret.txt
mv secret.txt ~/ctf-machine/stego/
mv resume.pdf ~/ctf-machine/stego/
ls ~/ctf-machine/stego
image5.jpg resume.pdf secret.txt
nano ~/ctf-machine/stego/secret.txt
sudo nano /var/www/html/hidden.html
mv ~/ctf-machine/stego/resume.pdf ~/ctf-machine/stego/
mv: '/root/ctf-machine/stego/resume.pdf' and '/root/ctf-machine/stego/resume.pdf' are the same file
ls ~/ctf-machine/stego
image5.jpg resume.pdf secret.txt
KALI
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



11. Credential Discovery Process

Users were required to perform enumeration on the web server and analyze steganographic output to retrieve valid SSH credentials.



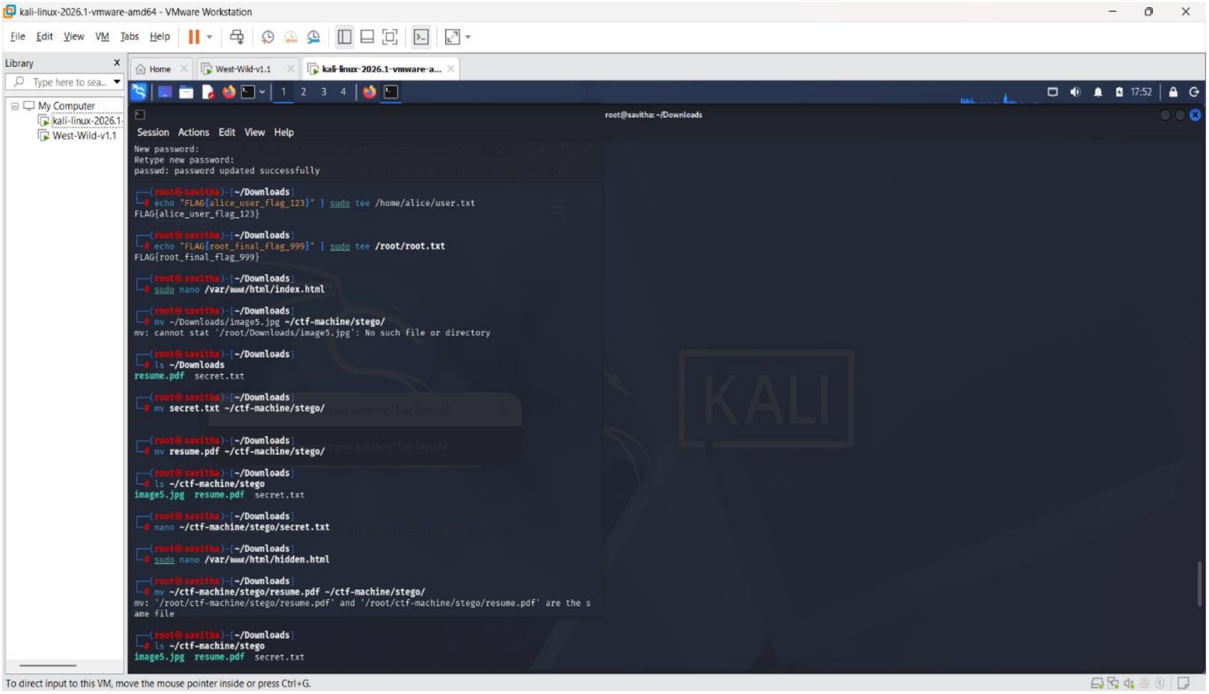


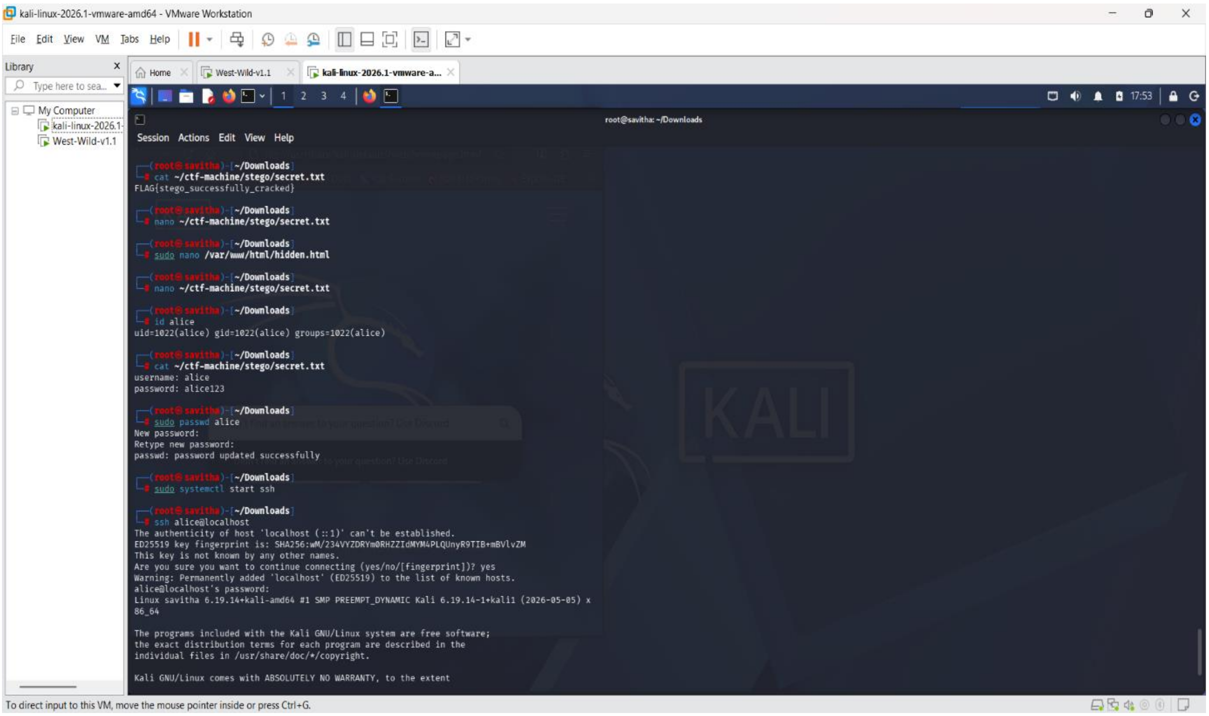
12. SSH Access

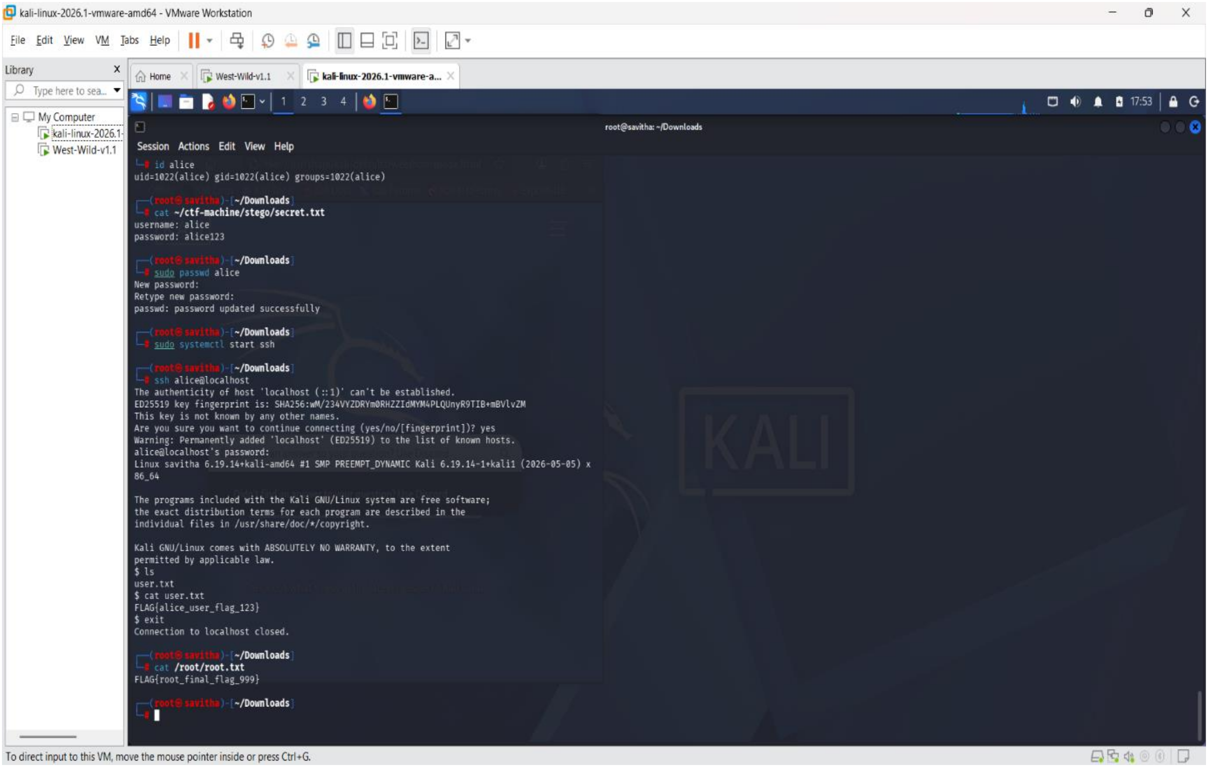
SSH service was used as the primary access mechanism after credential discovery. The attacker logs in as user alice to gain initial system access.

13. Flags

User Flag Location: /home/alice/user.txt → FLAG{alice_user_flag_123}
Root Flag Location: /root/root.txt → FLAG{root_final_flag_999}


kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
kali-linux-2026.1
West-Wild-v1.1
Home
West-Wild-v1.1
kali-linux-2026.1-vmware-a...
17:52
root@savitha: ~/Downloads
Session Actions Edit View Help
New password:
Retype new password:
passwd: password updated successfully
(root㉿savitha)-[~/Downloads]
echo "FLAG{alice_user_flag_123}" | sudo tee /home/alice/user.txt
FLAG{alice_user_flag_123}
(root㉿savitha)-[~/Downloads]
echo "FLAG{root_final_flag_999}" | sudo tee /root/root.txt
FLAG{root_final_flag_999}
(root㉿savitha)-[~/Downloads]
sudo nano /var/www/html/index.html
(root㉿savitha)-[~/Downloads]
mv ~/Downloads/image5.jpg ~/ctf-machine/stego/
mv: cannot stat '/root/Downloads/image5.jpg': No such file or directory
(root㉿savitha)-[~/Downloads]
ls ~/Downloads
resume.pdf secret.txt
(root㉿savitha)-[~/Downloads]
mv secret.txt ~/ctf-machine/stego/
(root㉿savitha)-[~/Downloads]
mv resume.pdf ~/ctf-machine/stego/
(root㉿savitha)-[~/Downloads]
ls ~/ctf-machine/stego
image5.jpg resume.pdf secret.txt
(root㉿savitha)-[~/Downloads]
nano ~/ctf-machine/stego/secret.txt
(root㉿savitha)-[~/Downloads]
sudo nano /var/www/html/hidden.html
(root㉿savitha)-[~/Downloads]
mv ~/ctf-machine/stego/resume.pdf ~/ctf-machine/stego/
mv: '/root/ctf-machine/stego/resume.pdf' and '/root/ctf-machine/stego/resume.pdf' are the same file
(root㉿savitha)-[~/Downloads]
ls ~/ctf-machine/stego
image5.jpg resume.pdf secret.txt
KALI
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


kali-linux-2026.1-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
kali-linux-2026.1
West-Wild-v1.1
Home
West-Wild-v1.1
kali-linux-2026.1-vmware-a...
17:53
root@savitha: ~/Downloads
Session Actions Edit View Help
(root㉿savitha)-[~/Downloads]
cat ~/ctf-machine/stego/secret.txt
FLAG{stego_successfully_cracked}
(root㉿savitha)-[~/Downloads]
nano ~/ctf-machine/stego/secret.txt
(root㉿savitha)-[~/Downloads]
sudo nano /var/www/html/hidden.html
(root㉿savitha)-[~/Downloads]
nano ~/ctf-machine/stego/secret.txt
(root㉿savitha)-[~/Downloads]
# id alice
uid=1022(alice) gid=1022(alice) groups=1022(alice)
(root㉿savitha)-[~/Downloads]
cat ~/ctf-machine/stego/secret.txt
username: alice
password: alice123
(root㉿savitha)-[~/Downloads]
sudo passwd alice
New password:
Retype new password:
passwd: password updated successfully
(root㉿savitha)-[~/Downloads]
sudo systemctl start ssh
(root㉿savitha)-[~/Downloads]
ssh alice@localhost
The authenticity of host 'localhost (::1)' can't be established.
ED25519 key fingerprint is: SHA256:wM/234VYZDRYm0RHZZIdMYM4PLQUnyR9TIB+mBVlvZM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
alice@localhost's password:
Linux savitha 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64
The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
KALI
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



14. Attack Flow

Web Enumeration → Hidden HTML Clues → Steganography Extraction → Credential Discovery → SSH Login → User Flag → Root Flag

15. Security Observations

The system demonstrates common vulnerabilities such as information disclosure, weak credential storage, and insecure hidden file exposure used intentionally for educational purposes.

16. Conclusion

The CTF machine successfully simulates a real-world penetration testing environment. It demonstrates multi-stage exploitation techniques and fulfills all academic requirements for cybersecurity training.